

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 3.2 PROBABILITY I: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Answer all five questions in the space under each prompt. Remember the puzzle we began with: during the 2023 Kenyan Premier League season, fans said Gor Mahia FC was 'due' for a loss after a winning streak, while betting advertisements on Nairobi matatus and social media told youth they could 'beat the odds'. Use what you learned in Lessons 1 to 8: your coin, die and token experiments, the probability formula, the range of probability, mutually exclusive events and independent events. Show all working, write probabilities as fractions in their simplest form, and explain your reasoning in full sentences. Where a question says 'suppose', use the numbers given for that question only.

Part 1: Does the past change the chance? (Equally likely events)

In our class experiment, each pair tossed a 10-shilling coin 40 times and we pooled the results. A fan at a Nairobi bus stage says: 'A coin has landed on heads five times in a row, so tails is now more likely.'

- (a) What is the theoretical probability of heads on one toss of a fair coin? Show how you get it using the formula.
- (b) Is the fan correct? Explain your answer using what you know about how one toss relates to the next.
- (c) Why did pooling the whole class's results give a more reliable estimate of the probability than your own pair's 40 tosses?
- (d) Name the mistake in thinking that the fan, and the fans who said Gor Mahia was 'due' for a loss, are making.

Write your answer here:

Part 2: Computing probabilities from the mandazi bag (Equally likely outcomes and the range of probability)

A cloth bag at a school canteen holds 15 tokens that represent mandazi flavours: 6 plain (white), 5 coconut (yellow) and 4 cardamom (green). One token is drawn at random.

- (a) Calculate $P(\text{plain})$, $P(\text{coconut})$ and $P(\text{cardamom})$ as fractions in simplest form.
- (b) Add the three probabilities. What does your total tell you about the possible outcomes of one draw?
- (c) Write down one event for this bag that has probability 0, and one that has probability 1.
- (d) A betting advertisement claims '120% chance of winning'. Using the range of probability, explain whether this claim can be true.

Write your answer here:

Part 3: Mutually exclusive events and the addition rule

- (a) Use the mandazi bag from Part 2. Are the events 'draw a plain mandazi' and 'draw a coconut mandazi' mutually exclusive on a single draw? Explain what mutually exclusive means.
- (b) Calculate $P(\text{plain or coconut})$ for one draw and show your rule.
- (c) In our token experiment each group drew 20 times with replacement. Why might a group's experimental result for 'plain or coconut' be close to, but not exactly equal to, your answer in (b)?
- (d) Suppose that, for one Gor Mahia match, we assume $P(\text{Win}) = 1/3$ and $P(\text{Draw}) = 1/3$ and that Win, Draw and Loss are the only possible results. (Real matches need not have equal chances; use these numbers for this question only.) Calculate $P(\text{Win or Draw})$. Then explain why a betting advertisement that only talks about 'Win or Draw' could mislead a bettor, and state when the addition rule must not be used.

Write your answer here:

Part 4: Independent events, the multiplication rule and why the stall always wins

- (a) Suppose that for each match the probability that Gor Mahia wins is $\frac{1}{3}$ (an assumed value for this question), and each match is independent. Calculate the probability that Gor Mahia wins two matches in a row, and then three matches in a row. Show the rule you use.
- (b) Gor Mahia has just won five matches in a row. A fan says the probability of winning the sixth match must now be different from $\frac{1}{3}$. Use the idea of independent events to respond.
- (c) A betting stall in Nairobi charges Ksh 20 per spin. A player wins Ksh 30 whenever the spinner lands on the winning sector, and $P(\text{win}) = \frac{1}{3}$ on each spin. Over 180 spins, a win is expected on about $\frac{1}{3}$ of the spins. Using this, work out the money you would expect the stall to collect, the money you would expect it to pay out, and the profit you would expect. Explain what this shows about gambling.

Write your answer here:

Part 5: Answering the Driving Question

Driving question: If past outcomes of a chance event, like a coin toss or a football match, do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?

Write a full explanation (about 8 to 10 sentences) that:

- uses all four ideas from this unit: equally likely events, the range of probability, mutually exclusive events and independent events;
- refers to the Gor Mahia winning-streak debate and the betting advertisements on Nairobi matatus and social media;
- ends by saying how this knowledge can help you make a responsible decision.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of probability calculations (formula, addition rule, multiplication rule)	All calculations are correct and in simplest form, the correct rule is named and used each time, and working is clearly shown.	Most calculations are correct with the correct rule used; there may be one small arithmetic slip or missing step of working.	Several calculations are incorrect, or rules are mixed up (for example adding when events should be multiplied), or working is missing.
Understanding of independence and the gambler's fallacy	Clearly explains that each independent event has a fixed probability that past results cannot change, names the gambler's fallacy, and applies it to both the coin and Gor Mahia.	States that past outcomes do not change future probability and applies it to the coin or the football example, but the explanation is less complete.	Does not reject the idea that a team or coin is 'due', or gives an explanation that is unclear or incorrect.
Understanding of the range of probability and mutually exclusive events	Correctly explains that $0 \leq P \leq 1$, that probabilities of all outcomes sum to 1, and when events are mutually exclusive and when the addition rule can and cannot be used.	Shows the range and the addition rule correctly but gives limited explanation of when the rule does not apply.	Confuses the range of probability or the meaning of mutually exclusive, or cannot say when the addition rule applies.
Connection to the Driving Question and the Kenyan context	Combines all four concepts in a coherent argument, explains how independence is compatible with predicting what is likely, uses the Gor Mahia debate, betting advertisements and the stall data	Uses most concepts and the Kenyan context correctly, and gives a reasonable conclusion about responsible decisions.	Gives a general answer with few concepts, little or no reference to the Kenyan phenomenon, or no conclusion about decision-making.

	accurately, and ends with a thoughtful responsible decision.		
Communication and reasoning	Writing is clear, logical and uses correct mathematical vocabulary in full sentences throughout.	Writing is mostly clear with mostly correct vocabulary.	Writing is hard to follow or key vocabulary is missing or misused.

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